

PARTHKUMAR PATEL

Software Engineer | Full-Stack Development | React, TypeScript, FastAPI, AI Applications

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EXPERIENCE

Software Development Intern | EventEase -Remote / NJ

Sep 2025 – Dec 2025

- Delivered two greenfield React 19 + TypeScript SPAs and extended a live tournament-scoring platform, building responsive interfaces optimized across mobile, desktop, and 4K TV displays.
- Architected modular frontend systems using Redux Toolkit and reusable shadcn/ui components, enabling shared component libraries across multiple applications.
- Integrated end-to-end REST APIs for authentication, scoring, leaderboards, and media uploads using Axios, implementing centralized error handling and user feedback mechanisms.
- Implemented near-real-time leaderboard synchronization through optimized polling across web and TV clients; collaborated through Git-based code reviews and pull requests.

Software Developer | TechBilv Solutions LLP -Gandhinagar, India

Jan 2023 – Aug 2024

- Enhanced client-facing React applications by rebuilding core UI components and navigation flows, improving usability and maintainability.
- Managed production deployments to AWS-backed IIS servers and configured static site hosting on AWS S3.
- Embedded a third-party AI chatbot and partnered cross-functionally with design and client stakeholders to deliver features on schedule.

PROJECTS

Refund Agent (LLM Tool-Use Governance) | Next.js, TypeScript, Anthropic Messages API | [Demo](#)

- Built an LLM customer-support refund agent (Next.js, TypeScript, Anthropic Messages API) with a hand-written streaming agentic loop and four tools, including a confusable refund/store-credit pair.
- Made policy enforcement structurally impossible to bypass by routing every model-requested side effect through a single rule gate, proven prompt-independent by a test blocking a \$900 refund with no system prompt.
- Wrote a trace-first eval suite of 9 scenarios, 46 assertions, and 43 unit tests runnable against a live, scripted, or recorded model, keeping CI free, offline, and deterministic.

Ledger (Hisab) - Shared Expense PWA | React, TypeScript, Supabase, TanStack Query | [Demo](#)

- Designed and shipped a mobile-first, installable PWA (React 19, TypeScript, Supabase Postgres/Auth/Realtime) that replaced a 3-person household's shared expense spreadsheet, live at \$0/month infra cost.
- Implemented real-time multi-device sync with optimistic mutations and rollback, so an entry logged on one phone appears on the others in under two seconds.
- Drove the build test-first with 193 Vitest tests, 48 pgTAP assertions against real Postgres, and 10 Playwright E2E specs - production Lighthouse 90 performance, 100 accessibility, 100 best practices.

MRTD Validation System (ICAO TD3) | Python, Automated Testing | [GitHub](#)

- Developed an end-to-end MRZ encode/decode and validation engine with Damm-based check digits and structured mismatch reporting.
- Authored 15 automated unit tests with a 100% pass rate and achieved 86% line coverage, above the 80% target.
- Executed mutation testing (406 mutants, 137 killed) and benchmarked performance at ~31.5K encodes/sec and ~156K decodes/sec on up to 10,000 records.

SKILLS

Languages: JavaScript, TypeScript, Python, Java, SQL, C++

Frontend: React, Next.js, Redux Toolkit, HTML5, CSS3, Tailwind CSS, shadcn/ui, Vite

Backend & Data: FastAPI, Node.js, Express.js, REST APIs, WebSocket, PostgreSQL, MongoDB

Cloud & Tools: AWS (S3), Git, GitHub, CI/CD

Testing & Practices: React Testing Library, Unit Testing, Agile/Scrum, Code Reviews, REST API Design, Responsive UI, Accessibility, OOP, Data Structures & Algorithms, LLM Integration

EDUCATION

Stevens Institute of Technology -Hoboken, NJ

Sep 2024 – May 2026

Master of Science in Computer Science | GPA: 3.9/4.0

Government Engineering College, Gandhinagar -India

Jun 2019 – Jul 2023

Bachelor of Engineering in Computer Engineering | CGPA: 8.33/10.0

CERTIFICATIONS

- Claude Code in Action, Building with the Claude API, Intro to Model Context Protocol** -Anthropic
- Advanced Prompt Engineering** -LinkedIn Learning